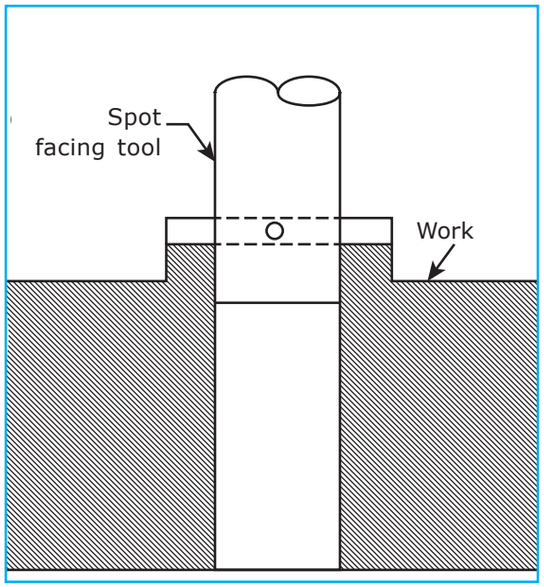
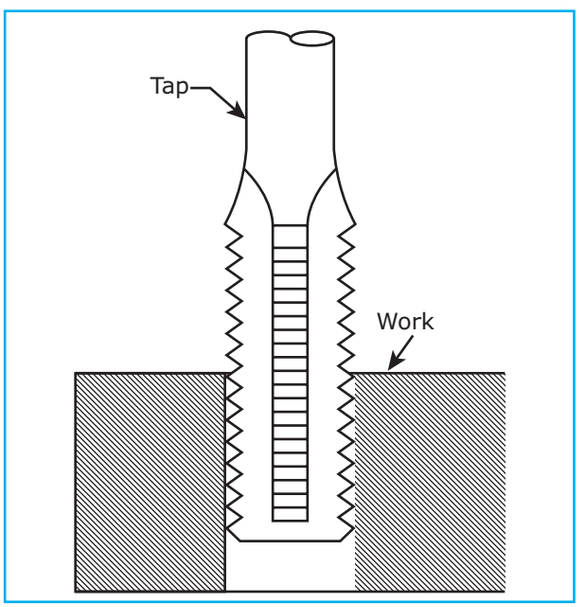


Spot Facing

Tapping

Tapping is the operation of cutting internal threads by means of a cutting tool called "Tap". Tapping in a drilling machine may be performed by hand (or) by power. When the tap is screwed into the hole. It removes metal and cuts internal threads which fit into external threads of the same size.

The following figure illustrates tapping operation.



Tapping

Calculation of the Tap drill size

Tap drill size may be derived from the following formula.

Tap drill size 'D'= T-2d

Where 'T' is the outer diameter of tap to be used. And 'd' is depth of the thread.

Tap drill size can also be calculated the following formula.

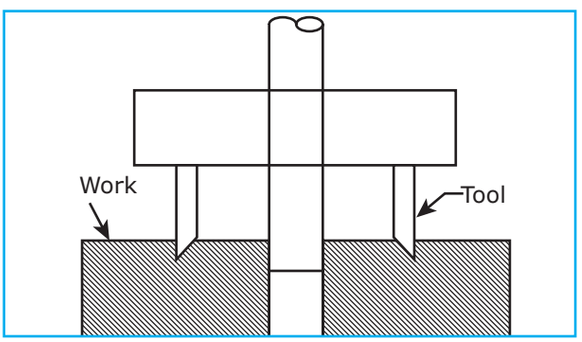
D=0.8T

Example:

Calculate the tap drill size. When outside diameter of the tap is 10mm the pitch of the thread is 1.5 mm and depth is 0.61mm.

Trepanning

Trepanning is the operation of producing a hole in sheet metal by removing metal along the circumference of a hollow cutting tool. Trepanning operation is performed for producing large holes. Fever chips are removed and much of the material is saved while the hole is produced. The tool may be operated at higher speeds. The speed depends upon the diameter of the hole to be made. The tool resembles a hollow tube having cutting edges at one end and a solid shank at the other to fit into the drill spindle.



Trepanning

The other operations can be performed in drilling machine are lapping and grinding.

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